System, energy, and flavor dependence of jets through di-hadron correlations in heavy ion collisions

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May 2009

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Abstract

QCD predicts a phase transition in nuclear matter at high energy densities. This matter, called Quark Gluon Plasma (QGP), should have very different properties from normal nuclear matter due to the high temperature and densities. This dense, hot nuclear matter should have much more in common with the matter created during the Big Bang than nuclear matter at normal densities. The Relativistic Heavy Ion Collider (RHIC) was built to study the QGP. Jets can act as a calibrated probe of a QGP, however,reconstruction of jets in a heavy ion environment is difficult. Therefore jets have been studied in heavy ion collisions by looking at the spacial correlations between two high-p_T hadrons in an event.

Di-hadron correlations provide a means to study jet production and fragmentation in heavy ion collisions. Previous studies have shown that the near-side di-hadron correlation peak can be decomposed into two components, the *Jet* and the *Ridge*. The *Jet* is narrow in both azimuth and pseudorapidity, while the *Ridge* is narrow in azimuth but independent of pseudorapidity within STAR's acceptance. STAR's data from Cu+Cu and Au+Au collisions at $\sqrt{s_{NN}} = 200$ GeV and 62 GeV allow comparative studies of these components in different systems and at different energies.

Data on correlations with both identified trigger particles and identified associated particles, including the first studies of identified associated particles in Cu+Cu and the energy dependence of these correlations are presented. The yields are studied as a function of N_{part}, $p_T^{trigger}$, and $p_T^{associated}$. The data indicate that the *Jet* component in heavy ion collisions is roughly independent of system size. Pythia simulations indicate that the $p_T^{trigger}$, $p_T^{associated}$, and collision energy dependence can be explained as kinematic effects expected if the *Jet* is dominantly formed by vacuum fragmentation.

The *Ridge* component is not present in p+p or d+Au collisions and could be an indication of modification of the *Jet* by the medium. The *Ridge* is consistent between systems for the same N_{part} and seems to have properties similar to the bulk. The data also show no strong dependence on the strangeness content of the trigger particle, including the Λ, K_S^0, and Ξ. Attempts have been made to explain the *Ridge* component, such as recombination, from momentum kicks, and from a plasma instability. Comparisons between the expectations of these models and the data are discussed.